

Student's name

Student's number

Teacher's name



PLC PRESBYTERIAN
LADIES' COLLEGE
SYDNEY
1888

2014
TRIAL
HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics

General Instructions

- Reading time - 5 minutes
- Working time - 3 hours
- Write using blue or black pen
Black is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total Marks – 100

Section I: Pages 3-6 10 marks

- Attempt questions 1-10, using the answer sheet on page 17.
- Allow about 15 minutes for this section

Section II: Pages 7-14 90 marks

- Attempt questions 11-16, using the lined paper provided.
- Allow about 2 hours 45 minutes for this section

Multiple Choice	11	12	13	14	15	16	Total
							%

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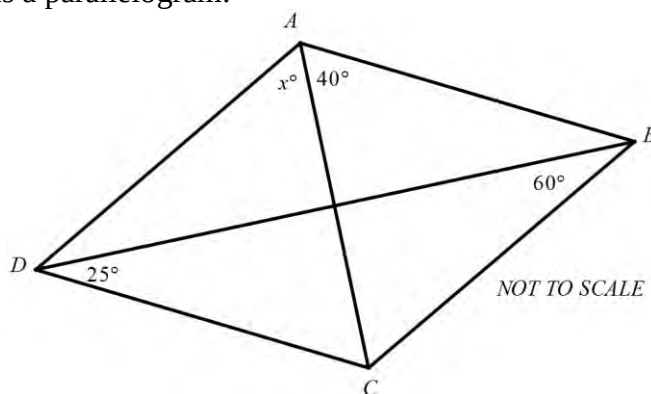
Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

1. What is the equation of the straight line through $(4, 5)$ and parallel to $2x - y + 1 = 0$?
- (A) $2x - y - 13 = 0$
(B) $2x - y - 3 = 0$
(C) $x + 2y - 10 = 0$
(D) $x + 2y - 7 = 0$
2. What are the nature of the roots of the quadratic equation $x^2 - 5x - 6 = 0$?
- (A) Real, rational and unequal
(B) Real, irrational and unequal
(C) Unreal, rational and unequal
(D) Unreal, irrational and unequal
3. $ABCD$ is a parallelogram.



What is the value of x ?

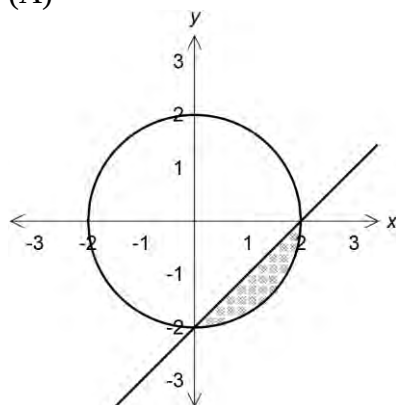
- (A) 40
(B) 45
(C) 55
(D) 60

4. What is the equation of the directrix of the parabola $y^2 = -16(x - 2)$?

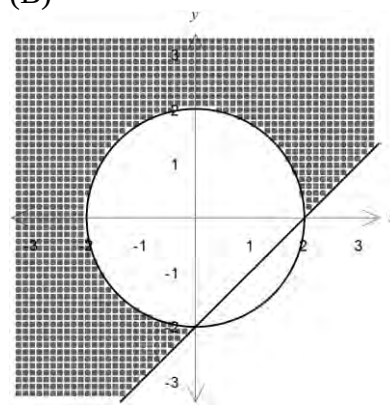
- (A) $x = -2$
 (B) $x = 6$
 (C) $y = -2$
 (D) $y = 6$

5. Which of the following diagrams show where $x^2 + y^2 \geq 4$ and $y \leq x - 2$ hold simultaneously?

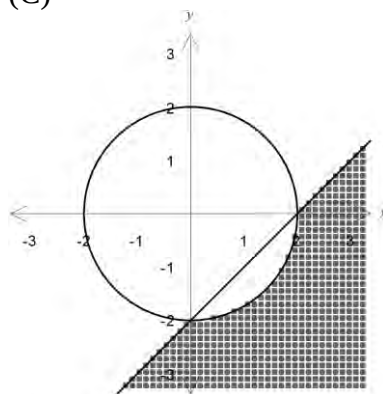
(A)



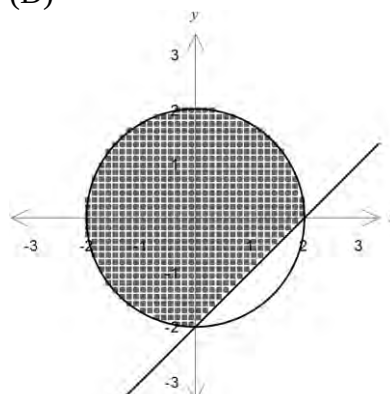
(B)



(C)



(D)



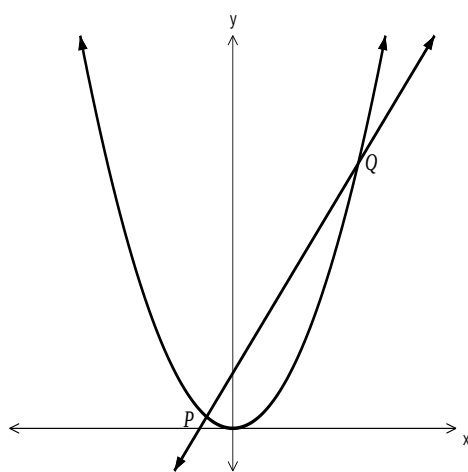
6. If $\log_a 3 = -1.585$ and $\log_a 5 = -2.322$, what is the value of $\log_a \left(\frac{27a}{5} \right)$?

- (A) -10.943
 (B) -1.433
 (C) $2.048a$
 (D) $6.143a$

7. What is the equation of the curve that passes through $(4, 5)$ if the gradient function is $\sqrt{2x+1}$?

- (A) $y = \frac{1}{3}(2x+1)^{\frac{3}{2}} - 4$
 (B) $x - 3y + 11 = 0$
 (C) $3x - y - 7 = 0$
 (D) $y = \frac{2}{3}(2x+1)^{\frac{3}{2}} - 13$

8. The x values of P and Q are the solutions to which quadratic equation?

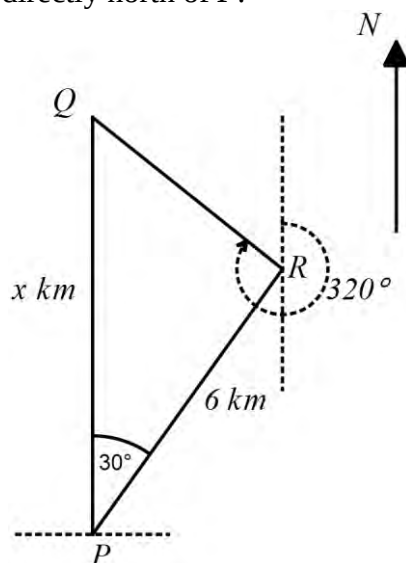


- (A) $x^2 - 3x - 3 = 0$
 (B) $x^2 - 3x + 3 = 0$
 (C) $x^2 + 3x - 3 = 0$
 (D) $x^2 + 3x + 3 = 0$
9. If $\sin x = -\frac{1}{5}$ and $\pi \leq x \leq \frac{3\pi}{2}$, then $\cot x$ equals

- (A) $-\frac{1}{2\sqrt{6}}$
 (B) $-2\sqrt{6}$
 (C) $\frac{1}{2\sqrt{6}}$
 (D) $2\sqrt{6}$

10.

A ship leaves a port, P , and sails 6 km on a bearing of 030° to position R . It then heads on a bearing of 320° until it reaches a port, Q , which is directly north of P .



Which of the following will give the value for x ?

- (A) $\frac{x}{\sin 30^\circ} = \frac{6}{\sin 110^\circ}$
- (B) $\frac{x}{\sin 40^\circ} = \frac{6}{\sin 110^\circ}$
- (C) $\frac{x}{\sin 110^\circ} = \frac{6}{\sin 30^\circ}$
- (D) $\frac{x}{\sin 110^\circ} = \frac{6}{\sin 40^\circ}$

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours 45 minutes for this section

Question 11 (15 marks) Use a SEPARATE writing booklet.

- a) Solve for x : 2
 $|3 - 2x| = 2x$
- b) Evaluate a and b if $\frac{2 - \sqrt{5}}{2 + \sqrt{5}} = a + \sqrt{b}$. 2
- c) Find $\lim_{x \rightarrow 0} \left[\frac{x^2 - 9x}{5x} \right]$ 1
- d) Find the domain of $y = \sqrt{3 - 2x - x^2}$. 2
- e) Differentiate the following with respect to x .
- (i) $\frac{e^{x^3}}{5x}$ 2
- (ii) $\log_e (x^2 + 1)^{\frac{1}{2}}$ 2
- f) Find $\int \frac{3 - x^2}{x} dx$ 2
- g) Show that $\int_0^{\log_e 2} \frac{2e^{2x}}{e^{2x} + 1} dx = \log_e \left(\frac{5}{2} \right)$. 2

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

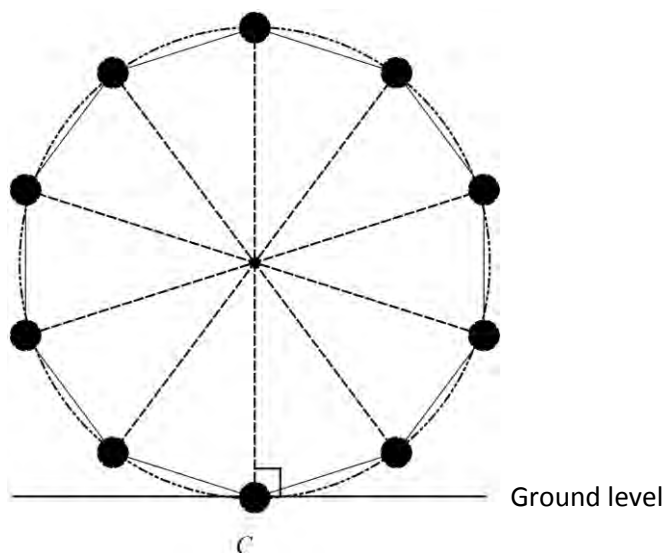
- a) Find the shortest distance between the lines $3x - 5y = 8$ and $3x - 5y = -1$. 2
- b) If $f(x) = x^3 - x + 4$, find the value of $f(f(-1))$. 2
- c) (i) Use Simpson's Rule with 3 function values to find an approximation to the area under the curve $y = \frac{1}{x}$ between $x = a$ and $x = 3a$ where a is positive. 2
- (ii) Hence show that $\log_e 3 \div \frac{10}{9}$. 1
- d) Sketch a continuous smooth curve for $x \geq 0$, where:
 $f(0) = 1$,
 $f'(x) < 0$ and $f''(x) > 0$ for $0 < x < 2$,
 $f'(2) = 0$,
 $f(2) = -2$
 $f'(x) > 0$ and $f''(x) > 0$ for $x > 2$. 2
- e) Prove $\sec \theta - \tan \theta - \frac{1}{\sec \theta - \tan \theta} = -2 \tan \theta$. 3
- f) Find the equation of the locus of the point $P(x, y)$ such that the distance from P to the point $A(2, 3)$ is twice the distance from P to the point $B(-1, 4)$. Write your answer in simplest form. 3

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

- a) Find the value(s) of p such that $x^2 + (p-1)x - (2p+1) > 0$ for all values of x . 2

- b) A Ferris Wheel has a radius of 40 metres and 10 cages. A particular cage, C , starts at ground level and travels on a circular path.



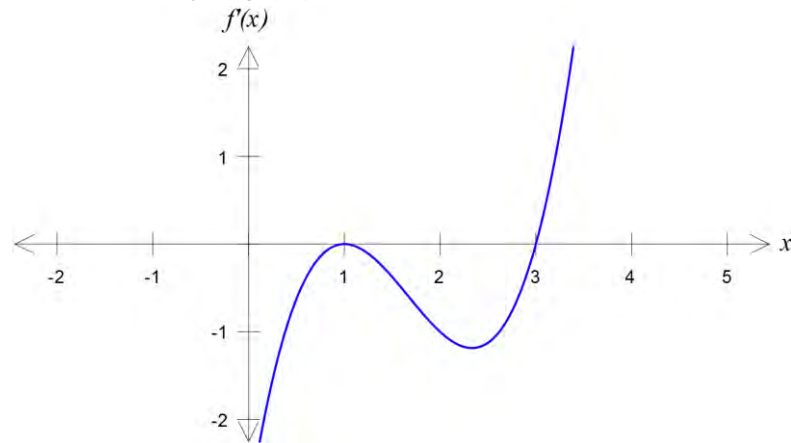
- (i) If the Ferris Wheel suddenly stops after cage C has moved 100 metres to a point D , through what angle has the wheel rotated? 1
- (ii) What is the area of the sector COD where O is the centre of the Ferris Wheel, C is the starting point of cage C and D is the point where cage C stopped? 1
- (iii) How far is cage C when it stops, in a straight line, from its starting point at ground level? Write your answer correct to 1 decimal place. 2
- (iv) What is its height, to the nearest metre, above the ground now? 2
- c) The region between $y = \sec x$ and the x -axis, bounded by $x = \frac{\pi}{6}$ and $x = \frac{\pi}{3}$ is rotated about the x -axis. Find the exact volume of the solid of revolution formed. 3

Question 13 continues on page 10

Question 13 continued

- d) If the roots of the equation $x^2 + ax + k = 0$ differ by $3a$, show that $k = -2a^2$ 2

- e) The graph of $y = f'(x)$ is shown below. 2



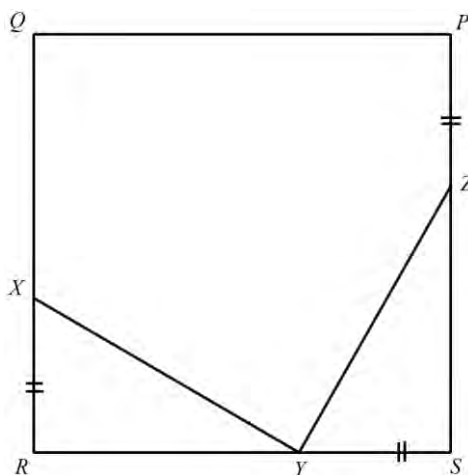
In your answer booklet, draw $y = f(x)$, clearly showing any stationary points.

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

- a)** A particle moving in a straight line with its velocity, v m/s at time t seconds is given by $v = 3\cos\left(2t - \frac{\pi}{2}\right)$.
- (i) Show that the particle is initially at rest? **1**
 - (ii) What is the maximum speed of the particle? **1**
 - (iii) Find the first time the velocity of the particle reaches $\frac{3\sqrt{3}}{2}$ m/s. **2**
 - (iv) Find the acceleration of the particle at time t . **1**
 - (v) Sketch the acceleration of the particle as a function of time for $0 \leq t \leq \pi$. **2**
 - (vi) If the particle is initially at the origin, find its distance travelled in the first π seconds. **3**

- b)** In the diagram PQRS is a square.



- (i) Prove that $\triangle XYR \equiv \triangle YZS$. **3**
- (ii) Prove that $\angle XYZ = 90^\circ$. **2**

End of Question 14

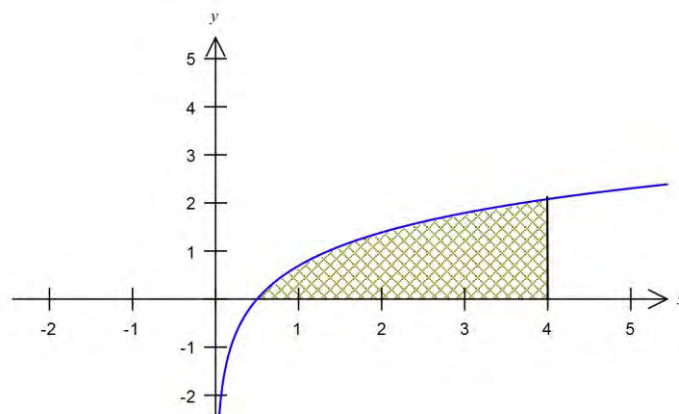
Question 15 (15 marks) Use a SEPARATE writing booklet.

- a)** The population of feral pigs is growing at a rate proportional to the current population. The population of pigs, P , at time t years is given by, $P = P_0 e^{kt}$, where P_0 and k are constants. In 2010, the feral pig population was first recorded. The population in 2012 was 350 and in 2014 it was 410.
- (i) Find the exact value of P_0 and k . 2
- (ii) What is the expected population of the pigs in 2020? 1
- (iii) In what year will the feral pig population reach 3000? 2
- b)** ABC is an isosceles triangle in which $a = b = 1\text{cm}$. $\angle C$ is obtuse. The perpendicular from B to AC produced, meets AC in D so that $BD = \frac{1}{2}AD$. Let $\angle BCD = \theta$. Show that $\sin \theta = \frac{1 + \cos \theta}{2}$ 2
- c)** For $y = 2x^2 e^x$
- (i) Find the x and y intercepts, if any. 1
- (ii) What happens to the function as $x \rightarrow -\infty$? 1
- (iii) Show that stationary points exist at $(0, 0)$ and $\left(-2, \frac{8}{e^2}\right)$ 2
- (iv) Determine the nature of the stationary points. 2
- (v) It is known that 2 points of inflexion exist on this curve at $x = -2 \pm \sqrt{2}$. Sketch the curve. 2

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

- a) (i) Show that the equation of the normal to the parabola $x^2 = 16y$ at the point where $x = 4$ is $2x + y - 9 = 0$. 2
- (ii) A line parallel to this normal is a tangent to the parabola. Find its equation and the co-ordinates of the point of contact. 3
- b) Find the area under the curve $y = \log_e 2x$, bounded by $x = 4$ and the x-axis. 3

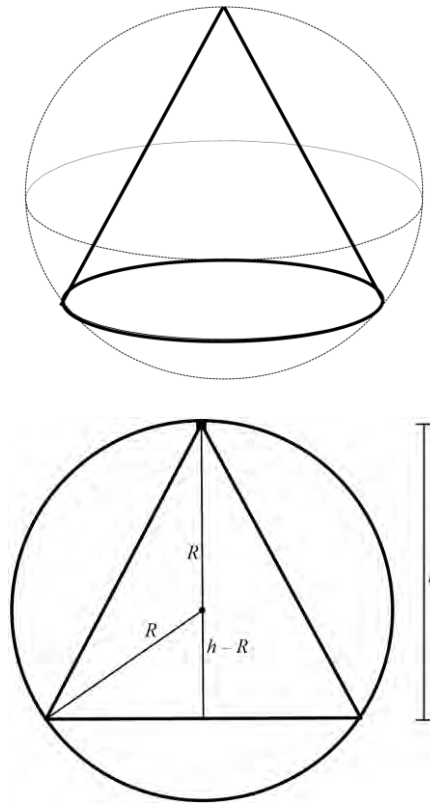


- c) A tap is slowly turned on such that the volume flow rate of water, R , varies with time according to the relation $R = kt$, where k is a constant and $t > 0$. 2
- Calculate the total volume of water that flows from the tap in the first 10 seconds if $k = 1.3 \text{ m}^3 / \text{s}^2$.

Question 16 continues on page 14

Question 16 continued

- d)** A right circular cone is inscribed in a sphere of radius R .



- (i) Show that the volume of the cone can be found by **2**
$$V = \frac{1}{3} \pi (2Rh - h^2) h$$
- (ii) Calculate the volume of the largest right circular cone inscribed in a sphere of radius R . Write your answer in terms of R . **3**

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

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Mathematics: Multiple Choice Answer Sheet

Student Number_____

Completely fill the response oval representing the most correct answer.

- | | | | | | | | | |
|-----|---|-----------------------|---|-----------------------|---|-----------------------|---|-----------------------|
| 1. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 2. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 3. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 4. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 5. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 6. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 7. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 8. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 9. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 10. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |

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Mathematics: Multiple Choice Answer Sheet

Student Number ANSWERS

Completely fill the response oval representing the most correct answer.

- | | | | | |
|-----|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| 1. | A ☐ | B ☒ | C ☐ | D ☐ |
| 2. | A ☒ | B ☐ | C ☐ | D ☐ |
| 3. | A ☐ | B ☐ | C ☒ | D ☐ |
| 4. | A ☐ | B ☒ | C ☐ | D ☐ |
| 5. | A ☐ | B ☐ | C ☒ | D ☐ |
| 6. | A ☐ | B ☒ | C ☐ | D ☐ |
| 7. | A ☒ | B ☐ | C ☐ | D ☐ |
| 8. | A ☒ | B ☐ | C ☐ | D ☐ |
| 9. | A ☐ | B ☐ | C ☐ | D ☒ |
| 10. | A ☐ | B ☐ | C ☐ | D ☒ |

Solutions for exams and assessment tasks

Academic Year	Yr 12	Calendar Year	2014
Course	2 Unit	Name of task/exam	Trials

Section I

1. $(4, 5)$ parallel to $2x - y + 1 = 0$

$$y = 2x + 1$$

$$\therefore m_1 = 2$$

$$m_2 = 2 \text{ since } m_1 = m_2 \text{ when parallel}$$

ieqn

$$y - y_1 = m(x - x_1)$$

$$y - 5 = 2(x - 4)$$

$$y - 5 = 2x - 8$$

$$2x - y - 3 = 0$$

$$\therefore B$$

2. $x^2 - 5x - 6 = 0$

$$\Delta = b^2 - 4ac$$

$$= (-5)^2 - 4(1)(-6)$$

$$= 25 + 24$$

$$= 49$$

$$\therefore \Delta > 0 \quad \begin{array}{l} \text{real roots} \\ \text{unequal roots} \end{array}$$

$$\Delta = 49 \text{ which is a perfect square}$$

$$\therefore \text{rational roots}$$

$$\therefore A$$

3. $\angle DCB = x + 40$ (opposite angles in parallelogram equal)

$$\therefore 25 + 60 + x + 40 = 180 \quad (\text{angle sum } \triangle DBC)$$

$$x = 55$$

$$\therefore C$$

4. $y^2 = -16(x - 2)$

of the form

$$(y - k)^2 = -4a(x - h)$$

$$\therefore \text{Vertex } (2, 0)$$

$$4a = 16$$

$$a = 4$$

opens to left

$$\therefore x = 6$$

$$\therefore B$$

5. $x^2 + y^2 \geq 4$

means region outside circle

$$\therefore B \text{ or } C$$

$$y \leq x - 2$$

$$\text{Test } (0, 0)$$

$$0 \not\leq -2$$

$$(0, 0) \text{ not in region}$$

$$\therefore C$$

$$\begin{aligned} 6. \log_a \left(\frac{27a}{5} \right) &= \log_a 27a - \log_a 5 \\ &= \log_a 27 + \log_a a - \log_a 5 \\ &= \log_a 3^3 + 1 - \log_a 5 \\ &= 3\log_a 3 + 1 - \log_a 5 \\ &= -1.433 \end{aligned}$$

$$\therefore B$$

Solutions for exams and assessment tasks

Academic Year	Yr 12	Calendar Year	2014
Course	2U	Name of task/exam	Trials

Q7. $(4, 5) \quad \frac{dy}{dx} = (2x+1)^{\frac{1}{2}}$

$$y = \int (2x+1)^{\frac{1}{2}} dx$$

$$y = \frac{(2x+1)^{\frac{3}{2}}}{\frac{3}{2}(2)} + c$$

$$y = \frac{(2x+1)^{\frac{3}{2}}}{3} + c$$

at $x=4 \quad y=5$

$$5 = \frac{9^{\frac{3}{2}}}{3} + c$$

$$5 = 9 + c$$

$$c = -4$$

$$\therefore y = \frac{1}{3} (2x+1)^{\frac{3}{2}} - 4$$

$\therefore A$

8. Parabola vertex $(0, 0)$

must have eqn $y = x^2$

line with positive gradient

and positive y-intercept

must have eqn $y = 3x + 3$

$\therefore$ solutions are the points of intersection

$$x^2 = 3x + 3$$

$$x^2 - 3x - 3 = 0$$

(This is the quadratic eqn)

$\therefore A$

9. $\sin x = -\frac{1}{5} \quad \pi \leq x \leq \frac{3\pi}{2}$

3rd quad

$$5^2 - (-1)^2 = a^2$$

$$a = \sqrt{24}$$

in 3rd quad

$$a = -\sqrt{24}$$

$$\therefore \cot x = \frac{\text{adj}}{\text{opp}}$$

$$= \frac{-\sqrt{24}}{-1}$$

$$= \sqrt{24}$$

$$= 2\sqrt{6}$$

$\therefore D$

10. bearing 320° has $180^\circ + \theta + \angle QRP$

$\angle \theta$ is alternate to $\angle QPR$

$$\therefore \angle \theta = 30^\circ$$

$$\therefore \angle QRP = 320 - 180 - 30 = 110^\circ$$

$$\therefore \frac{a}{\sin A} = \frac{b}{\sin B}$$

$$\frac{x}{\sin 110} = \frac{6}{\sin 40}$$

$\therefore D$

Solutions for exams and assessment tasks

Academic Year	Yr 12	Calendar Year	2014
Course	2 unit	Name of task/exam	Trials

Section II

Q 11 a $|3 - 2x| = 2x$

$$3 - 2x = 2x, \quad -3 + 2x = 2x$$

$$3 = 4x, \quad -3 \neq 0$$

$$x = \frac{3}{4}, \quad \text{no solution}$$

Check $x = \frac{3}{4}$

$$\text{LHS} = |3 - 2(\frac{3}{4})|$$

$$= |3 - \frac{3}{2}|$$

$$= \frac{3}{2}$$

$$\text{RHS} = 2(\frac{3}{4})$$

$$= \frac{3}{2}$$

$$\therefore \text{LHS} = \text{RHS}$$

$$\therefore x = \frac{3}{4} \text{ is solution}$$

b $\frac{2 - \sqrt{5}}{2 + \sqrt{5}} = a + \sqrt{b}$

$$\frac{(2 - \sqrt{5})}{(2 + \sqrt{5})} \times \frac{(2 - \sqrt{5})}{(2 - \sqrt{5})} = \frac{4 - 4\sqrt{5} + 5}{4 - 5}$$

$$= \frac{9 - 4\sqrt{5}}{-1}$$

$$= 4\sqrt{5} - 9$$

$$\therefore 4\sqrt{5} - 9 = a + \sqrt{b}$$

equating:

$$a = -9$$

$$b = 16 \times 5 = 80$$

c $\lim_{x \rightarrow 0} \frac{x^2 - 9x}{5x}$

$$= \lim_{x \rightarrow 0} \frac{x(x-9)}{5x}$$

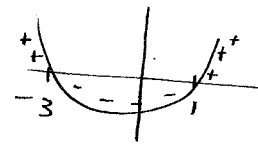
$$= \frac{-9}{5}$$

d $y = \sqrt{3 - 2x - x^2}$

$$\text{domain: } 3 - 2x - x^2 \geq 0$$

$$x^2 + 2x - 3 \leq 0$$

$$(x-1)(x+3) \leq 0$$



$$-3 \leq x \leq 1$$

e i) $\frac{d}{dx} \frac{e^{x^3}}{5x}$

$$= \frac{5x(3x^2 e^{x^3}) - e^{x^3}(5)}{(5x)^2}$$

ii) $\frac{d}{dx} \ln(x^2 + 1)^{\frac{1}{2}}$

$$= \frac{d}{dx} \frac{1}{2} \ln(x^2 + 1)$$

$$= \frac{1}{2} \frac{2x}{x^2 + 1}$$

$$= \frac{x}{x^2 + 1}$$

OR $\frac{\frac{1}{2}(x^2 + 1)^{-\frac{1}{2}}(2x)}{(x^2 + 1)^{\frac{1}{2}}}$

Solutions for exams and assessment tasks

Academic Year	Yr 12	Calendar Year	2014
Course	2 unit	Name of task/exam	Trials

$$\begin{aligned}
 \underline{f} \quad & \int \frac{3-x^2}{x} dx \\
 &= \int \left(\frac{3}{x} - x \right) dx \\
 &= \int \frac{3}{x} dx - \int x dx \\
 &= 3 \ln x - \frac{x^2}{2} + c
 \end{aligned}$$

$$\underline{g} \quad \text{RTS} \\
 \int_0^{\ln 2} \frac{2e^{2x}}{e^{2x}+1} dx = \ln\left(\frac{5}{2}\right)$$

$$\text{LHS} = \int_0^{\ln 2} \frac{2e^{2x}}{e^{2x}+1} dx$$

$$\text{since } \int \frac{f'(x)}{f(x)} dx = \ln f(x)$$

then

$$\begin{aligned}
 &= \left[\ln(e^{2x}+1) \right]_0^{\ln 2} \\
 &= \ln(e^{2\ln 2}+1) - \ln(e^0+1) \\
 &= \ln(e^{\ln 4}+1) - \ln(2) \\
 &= \ln(4+1) - \ln 2 \\
 &= \ln 5 - \ln 2 \\
 &= \ln\left(\frac{5}{2}\right)
 \end{aligned}$$

Q 12.

$$\underline{a} \quad 3x - 5y = 8$$

$$3x - 5y = -1$$

the shortest distance is the perpendicular distance.

A point on the line $3x - 5y = 8$ is $(1, -1)$

$\therefore$ perp. distance from this point to the other line is:

$$d_{\perp} = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$= \frac{|3(1) - 5(-1) + 1|}{\sqrt{3^2 + (-5)^2}}$$

$$= \frac{|3 + 5 + 1|}{\sqrt{34}}$$

$$= \frac{9}{\sqrt{34}}$$

$$= \frac{9\sqrt{34}}{34} \text{ units}$$

$$\underline{b} \quad f(x) = x^3 - x + 4$$

$$f(f(-1)) = f((-1)^3 - (-1) + 4)$$

$$= f(-1 + 1 + 4)$$

$$= f(4)$$

$$= 4^3 - 4 + 4$$

$$= 64$$

Solutions for exams and assessment tasks

Academic Year	Yr 12	Calendar Year	2014
Course	2 unit	Name of task/exam	Trials

$$c) i) A \div \frac{h}{3} [y_0 + y_n + 4y_1]$$

$$h = \frac{3a - a}{2} = a$$

$$\therefore A = \frac{a}{3} \left[\frac{1}{a} + \frac{1}{3a} + 4 \times \frac{1}{2a} \right]$$

$$= \frac{a}{3} \left[\frac{1}{a} + \frac{1}{3a} + \frac{2}{a} \right]$$

$$= \frac{a}{3} \left[\frac{3+1+6}{3a} \right]$$

$$= \frac{10}{3} \left(\frac{10}{3a} \right)$$

$$= \frac{10}{9}$$

$$ii) \int_a^{3a} \frac{1}{x} dx = [\ln x]_a^{3a}$$

$$= \ln 3a - \ln a$$

$$= \ln \left(\frac{3a}{a} \right)$$

$$= \ln 3$$

$$\therefore \ln 3 \div \frac{10}{9}$$

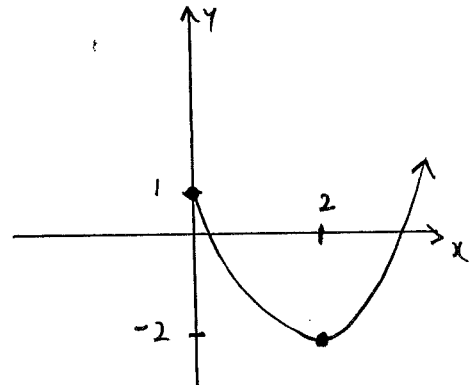
$$d) f(0)=1, f(2)=-2, f'(2)=0$$

$$f'(x) < 0 \text{ for } 0 < x < 2 \text{ decreasing}$$

$$f'(x) > 0 \text{ for } x > 2 \text{ increasing}$$

$$f''(x) > 0 \text{ for } 0 < x < 2 \text{ concave up}$$

$$f''(x) > 0 \text{ for } x > 2 \text{ concave up}$$



e) RTP

$$\sec \theta - \tan \theta - \frac{1}{\sec \theta - \tan \theta} = -2 \tan \theta$$

$$\text{LHS} = \frac{\sec \theta - \tan \theta}{1} - \frac{1}{\sec \theta - \tan \theta}$$

$$= \frac{(\sec \theta - \tan \theta)^2 - 1}{\sec \theta - \tan \theta}$$

$$= \frac{\sec^2 \theta - 2 \sec \theta \tan \theta + \tan^2 \theta - 1}{\sec \theta - \tan \theta}$$

$$= \frac{\tan^2 \theta + 1 - 2 \sec \theta \tan \theta + \tan^2 \theta - 1}{\sec \theta - \tan \theta}$$

$$= \frac{2 \tan \theta (\tan^2 \theta - \sec \theta)}{(\sec \theta - \tan \theta)}$$

(using identity
 $\tan^2 \theta + 1 = \sec^2 \theta$)

$$= -2 \tan \theta$$

$$= \text{RHS}$$

Solutions for exams and assessment tasks

Academic Year	Yr 12	Calendar Year	2014
Course	2 unit	Name of task/exam	Trials

$$f) d_{PA} = 2 d_{PB}$$

$$\sqrt{(x-2)^2 + (y-3)^2} = 2\sqrt{(x+1)^2 + (y-4)^2}$$

square both sides

$$(x-2)^2 + (y-3)^2 = 4[(x+1)^2 + (y-4)^2]$$

$$x^2 - 4x + 4 + y^2 - 6y + 9 = 4[x^2 + 2x + 1 + y^2 - 8y + 16]$$

$$\Rightarrow \underline{x^2 - 4x + y^2 - 6y + 13} = \underline{4x^2 + 8x + 4 + 4y^2 - 32y + 64}$$

$$3x^2 + 12x + 3y^2 - 26y + 55 = 0$$

Q13

$$a) x^2 + (p-1)x - (2p+1) > 0$$

 $\therefore$ positive def.

$$a=1 \quad a > 0$$

$$\Delta = (p-1)^2 - 4(1)(-2p-1)$$

$$= p^2 - 2p + 1 - 4(-2p-1)$$

$$= p^2 - 2p + 1 + 8p + 4$$

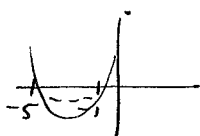
$$= p^2 + 6p + 5$$

$$\text{for pos. def } \Delta < 0$$

$$\therefore p^2 + 6p + 5 < 0$$

$$(p+1)(p+5) < 0$$

$$-5 < p < -1$$



$$b) i) l = r\theta$$

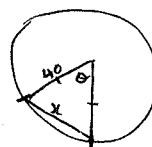
$$100 = 40\theta$$

$$\theta = \frac{100}{40} = \frac{10}{4} = \frac{5}{2} \text{ radians}$$

$$ii) A = \frac{1}{2} r^2 \theta$$

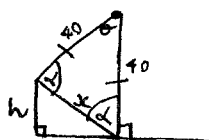
$$= \frac{1}{2} (40)^2 \left(\frac{5}{2}\right)$$

$$= 2000 \text{ m}^2$$

iiicalculator
in radians

$$x^2 = 40^2 + 40^2 - 2 \times 40 \times 40 \cos\left(\frac{5}{2}\right)$$

$$x = 75.9 \text{ m (1 dp)}$$

iv

$$\alpha = \frac{\pi - \theta}{2} = \frac{\pi - \frac{5}{2}}{2} \approx 0.32 \text{ rads}$$

$$\therefore \frac{\pi}{2} - 0.32 \dots = 1.25$$

$$\therefore \sin 1.25 = \frac{h}{x}$$

$$h = x \sin 1.25$$

$$= 75.9 \times \sin 1.25$$

$$= 72.0 \text{ m.}$$

$$= 72 \text{ m (nearest metre)}$$

Solutions for exams and assessment tasks

Academic Year	Yr 12	Calendar Year	2014
Course	2 unit	Name of task/exam	Trials

$$\begin{aligned}
 \underline{c} \quad V &= \pi \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} (\sec x)^2 dx \\
 &= \pi \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sec^2 x dx \\
 &= \pi \left[\tan x \right]_{\frac{\pi}{6}}^{\frac{\pi}{3}} \\
 &= \pi \left[\tan \frac{\pi}{3} - \tan \frac{\pi}{6} \right] \\
 &= \pi \left[\sqrt{3} - \frac{1}{\sqrt{3}} \right] \\
 &= \pi \left[\frac{3-1}{\sqrt{3}} \right] \\
 &= \frac{2\pi}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} \\
 &= \frac{2\sqrt{3}\pi}{3} \text{ units}^3
 \end{aligned}$$

$$\underline{d} \quad x^2 + ax + k = 0$$

if roots differ by $3a$

Let roots be α , β and $\alpha - \beta = 3a$

Sum of roots

$$\alpha + \beta = -a \quad (1)$$

Product of roots

$$\alpha\beta = k \quad (2)$$

also

$$\alpha - \beta = 3a \quad (3)$$

Solve (1) and (3)

$$2\alpha = 2a$$

$$\alpha = a$$

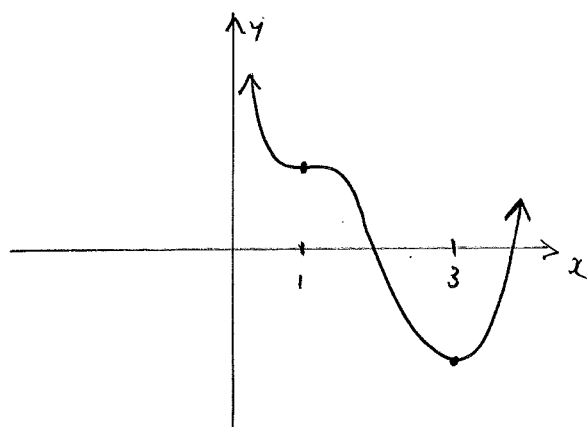
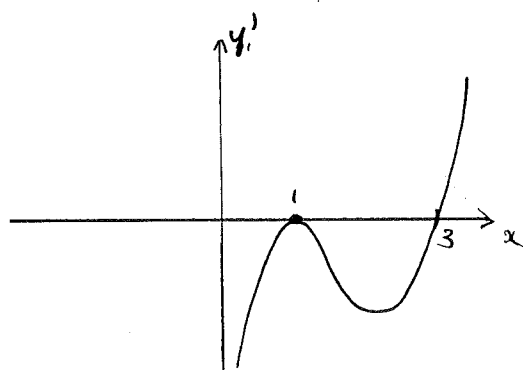
$$\therefore \beta = -2a$$

$$\therefore \alpha\beta = k \text{ and}$$

$$\alpha\beta = a(-2a)$$

$$\therefore k = -2a^2$$

e



when $y' = 0$ (at $x=1$)

$$\begin{array}{c|c|c|c}
 x & 0^- & 0 & 0^+ \\
 \hline
 y' & - & 0 & -
 \end{array}$$

$\therefore \text{local min}$

when $y' = 0$ (at $x=3$)

$$\begin{array}{c|c|c|c}
 x & 3^- & 3 & 3^+ \\
 \hline
 y' & - & 0 & +
 \end{array}$$

$\therefore \text{local min}$

Solutions for exams and assessment tasks

Academic Year	Yr 12	Calendar Year	2014
Course	2 unit	Name of task/exam	Trials

Q14

a $V = 3 \cos\left(2t - \frac{\pi}{2}\right)$

i if initially at rest then
when $t = 0$, $v = 0$.

Show $t = 0$

$$\begin{aligned} V &= 3 \cos\left(0 - \frac{\pi}{2}\right) \\ &= 3 \cos\left(-\frac{\pi}{2}\right) \\ &= 3(0) \\ &= 0 \end{aligned}$$

$\therefore$ initially at rest.

ii max speed is when
 $V = 3$.

Note: $\cos\left(2t - \frac{\pi}{2}\right)$ is 1
as its maximum, & -1
as its minimum

iii $\frac{3\sqrt{3}}{2} = 3 \cos\left(2t - \frac{\pi}{2}\right)$

$$\frac{\sqrt{3}}{2} = \cos\left(2t - \frac{\pi}{2}\right)$$

$$-\frac{\pi}{6}, \frac{\pi}{6}, \frac{5\pi}{6}, \dots = 2t - \frac{\pi}{2}$$

$$\frac{2\pi}{6}, \frac{4\pi}{6}, \frac{8\pi}{6}, \dots = 2t$$

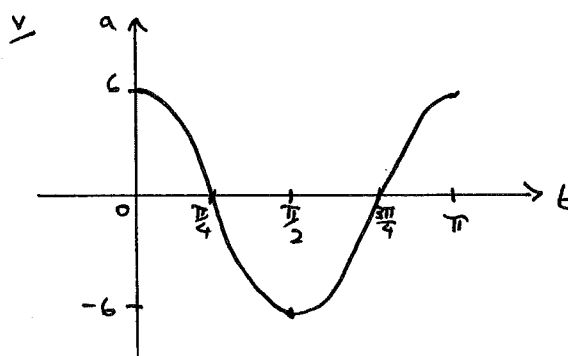
$$t = \frac{\pi}{6}, \frac{2\pi}{6}, \dots$$

$\therefore$ first time $V = \frac{3\sqrt{3}}{2}$ is at

$$t = \frac{\pi}{6}.$$

iv $a = -6 \sin\left(2t - \frac{\pi}{2}\right)$

$\downarrow$
S
C
-S
-C



vi $t = 0$ $x = 0$

$$V = 3 \cos\left(2t - \frac{\pi}{2}\right)$$

$$x = \frac{3 \sin\left(2t - \frac{\pi}{2}\right)}{2} + C$$

when $t = 0$ $x = 0$

$$0 = \frac{3}{2} \sin\left(0 - \frac{\pi}{2}\right) + C$$

$$0 = -\frac{3}{2} + C$$

$$C = \frac{3}{2}$$

$$\therefore x = \frac{3}{2} \sin\left(2t - \frac{\pi}{2}\right) + \frac{3}{2}$$

between $t = 0$ and $t = \pi$

particle turns around at $t = \frac{\pi}{2}$.

$$\therefore t = 0 \quad x = 0$$

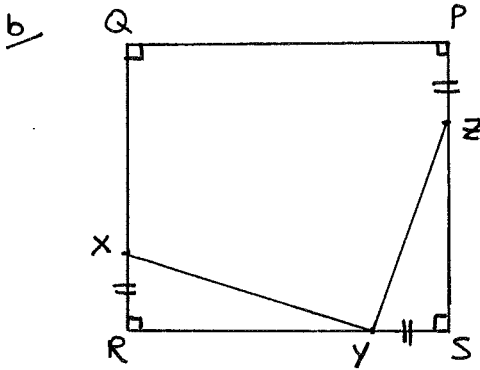
$$t = \frac{\pi}{2} \quad x = 3$$

$$t = \pi \quad x = 0$$

$\therefore$ total distance travelled in
first π seconds is 6 metres.

Solutions for exams and assessment tasks

Academic Year	Yr 12	Calendar Year	2014
Course	2 unit	Name of task/exam	Trials



i. In $\triangle XRY$ and $\triangle ZPS$

$$\angle XRY = \angle ZPS \quad (\text{all angles in square} = 90^\circ)$$

$$XR = YS \quad (\text{given})$$

$$RY + YS = ZS + ZP \quad (\text{all sides in square are equal})$$

$$\text{Since } YS = ZP \quad (\text{given})$$

$$RY = ZS$$

$$\therefore \triangle XRY \equiv \triangle ZPS \quad (\text{SAS})$$

ii. Let $\angle XRY = x$

$$\therefore \angle ZPS = x \quad (\text{in congruent triangles corresponding angles are equal})$$

$$\therefore \angle ZPS = 90 - x \quad (\text{angle sum } \triangle ZPS)$$

$$\therefore \angle XYZ = 180 - x - (90 - x) \quad (\text{angle sum of straight line})$$

$$\therefore \angle XYZ = 180 - x - 90 + x = 90$$

$$\therefore \angle XYZ = 90^\circ$$

Q15

$$a. i. P = P_0 e^{kt}$$

2010 records commences

$$t: 2012 \Rightarrow t = 2 \quad P = 350$$

$$t: 2014 \Rightarrow t = 4 \quad P = 410$$

$$\therefore 350 = P_0 e^{k \times 2}$$

$$410 = P_0 e^{k \times 4}$$

$$\therefore 350 = P_0 e^{2k} \quad (1)$$

$$410 = P_0 e^{4k} \quad (2)$$

$$\frac{410}{350} = \frac{P_0 e^{4k}}{P_0 e^{2k}} \quad (2) \div (1)$$

$$\frac{41}{35} = e^{4k - 2k}$$

$$\frac{41}{35} = e^{2k}$$

$$\ln\left(\frac{41}{35}\right) = 2k$$

$$k = \frac{1}{2} \ln\left(\frac{41}{35}\right)$$

$$\therefore 350 = P_0 e^{\frac{1}{2} \ln\left(\frac{41}{35}\right) \times 2}$$

$$P_0 = \frac{350 \times 35}{41} = \frac{12250}{41}$$

$$ii. t: 2020 \Rightarrow t = 10$$

$$P = 298.78... e^{\frac{1}{2} \ln\left(\frac{41}{35}\right) \times 10}$$

$$P = 659$$

$$iii. \quad 3000 = 298.78 e^{\frac{1}{2} \ln\left(\frac{41}{35}\right) t}$$

$$\frac{3000}{298.78...} = e^{\frac{1}{2} \ln\left(\frac{41}{35}\right) t}$$

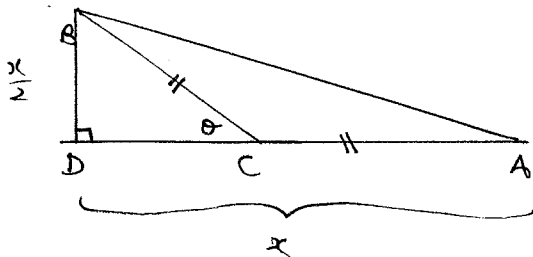
$$\ln\left(\frac{3000}{298.78...}\right) = \frac{1}{2} \ln\left(\frac{41}{35}\right) t$$

$$\therefore t = 29.2 \quad \therefore 2040$$

Solutions for exams and assessment tasks

Academic Year	Yr 12	Calendar Year	2014
Course	2 unit	Name of task/exam	Trials

b



$$\text{Let } BC = y$$

$$\therefore CA = y$$

$$\therefore \sin \theta = \frac{BD}{BC}$$

$$\sin \theta = \frac{x/2}{y}$$

$$\therefore y = \frac{x}{2 \sin \theta} \quad \therefore \frac{x}{y} = 2 \sin \theta$$

$$\begin{aligned} \cos \theta &= \frac{DC}{BC} \\ &= \frac{x - y}{y} \end{aligned}$$

$$\cos \theta = \frac{x}{y} - 1$$

$$\cos \theta = 2 \sin \theta - 1$$

$$\therefore \sin \theta = \frac{\cos \theta + 1}{2}$$

$$c \quad y = 2x^2 e^x$$

i for x-intercepts: set $y = 0$

$$\therefore 0 = 2x^2 e^x$$

$$\therefore x = 0, \quad e^x \neq 0$$

$$\therefore x = 0$$

for y-intercept: set $x = 0$

$$\therefore y = 0e^0$$

$$y = 0$$

$\therefore (0, 0)$ is x & y intercepts

ii as $x \rightarrow -\infty$

$$y \rightarrow \frac{2x^2}{e^x} \rightarrow 0^+$$

$$y \rightarrow 0$$

iii

$$\frac{dy}{dx} = uv' + vu'$$

$$u = 2x^2$$

$$u' = 4x$$

$$v = e^x$$

$$v' = e^x$$

$$= 2x^2 e^x + e^x (4x)$$

$$= e^x (2x^2 + 4x)$$

for stat pts $\frac{dy}{dx} = 0$

$$e^x (2x^2 + 4x) = 0$$

$$\therefore \underbrace{e^x}_{\text{no soln}} = 0 \quad 2x^2 + 4x = 0$$

$$2x(x+2) = 0$$

$$\therefore x = 0, \quad x = -2$$

$\therefore$ Stat pts at $(0, 0), (-2, \frac{8}{e^2})$

iv

x	$-\frac{1}{2}$	0	$\frac{1}{2}$
$\frac{dy}{dx}$	-0.9	0	4.1

+ $\searrow$ / +

$\therefore \min(0, 0)$

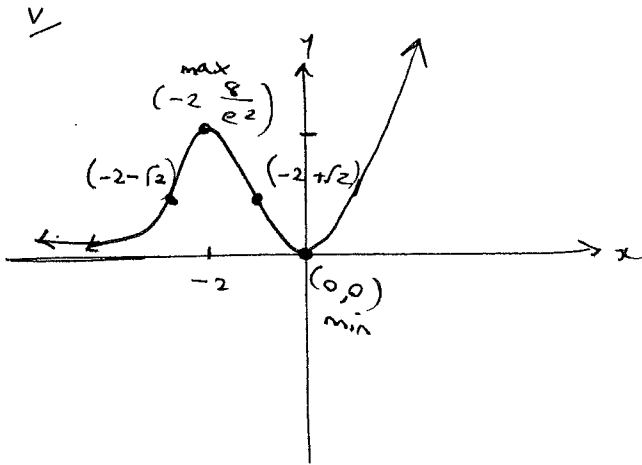
x	-3	-2	$-\frac{1}{2}$
$\frac{dy}{dx}$	0.3	0	-0.9

+ $\searrow$ -

$\therefore \max(-2, \frac{8}{e^2})$

Solutions for exams and assessment tasks

Academic Year	Yr 12	Calendar Year	2014
Course	2 unit	Name of task/exam	Trials

Q16

q. 1, $x^2 = 16y$

$$y = \frac{x^2}{16}$$

$$\frac{dy}{dx} = \frac{2x}{16}$$

at $x = 4$

$$m_{\text{tangent}} = \frac{8}{16} = \frac{1}{2}$$

$$\therefore m_{\text{norm}} = -2$$

at $x = 4$ $y = \frac{4^2}{16} = 1$

$\therefore (4, 1)$ is point

$\therefore$ eqn normal is

$$y - 1 = -2(x - 4)$$

$$y - 1 = -2x + 8$$

$$2x + y - 9 = 0$$

is eqn of normal.

ii $y = \frac{x^2}{16}$

$$\frac{dy}{dx} = \frac{2x}{16} = \frac{x}{8}$$

if parallel to normal $m = -2$

$$\frac{x}{8} = -2$$

$$\therefore x = -16$$

at $x = -16$

$$y = 16$$

$\therefore (-16, 16)$ are the coordinates of the point of contact.

The equation is:

$$y - 16 = -2(x + 16)$$

$$y - 16 = -2x - 32$$

$$2x + y + 16 = 0$$

b $A = \int_a^b y \, dx$

$$\begin{aligned} y &= \ln 2x \\ e^y &= 2x \\ x &= \frac{1}{2}e^y \end{aligned}$$

$$= \int_a^b \ln 2x \, dx$$

$$= \text{rectangle} - \int_0^{\ln 8} x \, dy$$

$$= 4 \times \ln 8 - \int_0^{\ln 8} \frac{1}{2} e^y \, dy$$

$$= 4 \ln 2^3 - \left[\frac{1}{2} e^y \right]_0^{\ln 8}$$

$$= 12 \ln 2 - \left(\frac{1}{2} e^{\ln 8} - \frac{1}{2} e^0 \right)$$

$$= 12 \ln 2 - \left(4 - \frac{1}{2} \right)$$

$$= 12 \ln 2 - \frac{7}{2}$$

Solutions for exams and assessment tasks

Academic Year	Yr 12	Calendar Year	2014
Course	Unit	Name of task/exam	Trials

$$c) \frac{dV}{dt} = kt$$

$$V = \int kt \, dt$$

$$V = \frac{kt^2}{2} + C$$

$$\text{when } t=0 \quad V=0 \quad \therefore C=0$$

$$\therefore V = \frac{1}{2} kt^2$$

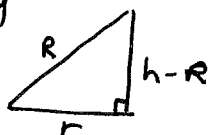
$$\text{when } k=1.3, \quad t=10$$

$$V = \frac{1}{2} \times 1.3 \times 10^2$$

$$V = 65 \text{ m}^3$$

$$d) \quad V_{\text{cone}} = \frac{1}{3} \pi r^2 h$$

radius of cone, r , is
found using



$$r^2 = R^2 - (h-R)^2$$

$$r^2 = R^2 - h^2 + 2hR - R^2$$

$$r^2 = 2hR - h^2$$

$$\therefore V_{\text{cone}} = \frac{1}{3} \pi (2hR - h^2) h$$

ii) largest cone when sphere
has radius R .

$\therefore R$ is constant
 h is variable

$$V = \frac{2}{3} \pi h^2 R - \frac{1}{3} \pi h^3$$

$$\frac{dV}{dh} = \frac{4}{3} \pi R h - \pi h^2$$

$$\text{for max/min } \frac{dV}{dh} = 0$$

$$\therefore h \left(\frac{4}{3} \pi R - \pi h \right) = 0$$

$$h=0, \quad \frac{4}{3} \pi R - \pi h = 0$$

$$\frac{4}{3} R = h$$

$$h = \frac{4R}{3}$$

$$\frac{d^2V}{dh^2} = \frac{4}{3} \pi R - 2\pi h$$

$$\text{when } h=0$$

$$\frac{d^2V}{dh^2} = \frac{4}{3} \pi R$$

$$> 0$$

$\cup \therefore \text{min}$

$$\text{when } h = \frac{4R}{3}$$

$$\frac{d^2V}{dh^2} = \frac{4}{3} \pi R - 2\pi \frac{4R}{3}$$

$$= -\frac{4\pi R}{3}$$

$$< 0$$

$\cap \therefore \text{max}$

$\therefore$ Largest cone is when

$$V = \frac{2}{3} \pi \left(\frac{4R}{3} \right)^2 R - \frac{1}{3} \pi \left(\frac{4R}{3} \right)^3$$

$$V = \frac{2}{3} \pi R \left(\frac{16R^2}{9} \right) - \frac{1}{3} \pi \left(\frac{64R^3}{27} \right)$$

$$V = \frac{32\pi R^3}{81} \quad \text{units}^3$$